

ALLWIN DONALD CHRISTEVE E

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PROFESSIONAL SUMMARY

Fresher Medical Electronics graduate (B.E., 2025) with hands-on experience in AI/ML-driven image processing, biomedical signal analysis (EEG), and Agentic AI applications. Skilled in Python, feature extraction, classification techniques, OpenAI API integration, and building AI-powered automation workflows. Seeking an entry-level AI/ML/Python role to apply problem-solving skills and real-world project experience.

TECHNICAL SKILLS

- Programming: Python (Intermediate), SQL (Basic)
- Agentic AI & LLMs: OpenAI API, GPT-3.5, Prompt Engineering, Agentic AI Workflows
- Framework: LangChain (basic workflows)
- Biomedical: EEG preprocessing, feature extraction, seizure/non-seizure classification (project-based)
- Image Processing: Adaptive Thresholding, Feature Extraction, Segmentation, RANSAC (project-based)

EDUCATION

Saveetha Engineering College	Chennai, India
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Bachelor of Engineering	May 2025
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Major in Medical Electronics

CGPA: 7.7

St.Joseph's Matriculation Higher Secondary School	Hosur, India
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Higher Secondary Certificate	2021
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Cumulative Percentage: 83%

St.Joseph's Matriculation Higher Secondary School	Hosur, India
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Secondary School Certificate	2019
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Cumulative Percentage: 82%

PROJECTS

AI-Based Intraocular Pressure (IOP) Estimation from Fundus Images

- Developed an AI-assisted image processing pipeline to estimate synthetic IOP indicators from retinal fundus images.
- Implemented ML-oriented feature extraction for optic disc and cup segmentation using RANSAC and adaptive thresholding.
- Extracted clinical features including CDR, rim ratio, vessel density, brightness, and contrast for data-driven estimation.
- Built a GUI to visualize AI-based feature extraction and analysis stages.

Agentic AI System for IT News Retrieval, Categorization, and Summarization using OpenAI API

- Developed an Agentic AI-based application to automatically retrieve real-time IT news using APIs.
- Integrated OpenAI API for intelligent news categorization and abstractive summarization.
- Focused on affordability and safety considerations for broader accessibility.

Machine Learning-Based Detection of Epileptic Seizures (MATLAB)

- Preprocessed EEG signals and removed artifacts for ML-based biomedical signal analysis.
- Extracted discriminative features and trained a supervised classifier to detect seizure vs non-seizure states.
- Designed autonomous AI agents for task orchestration including data fetching, classification, and summarization.
- Implemented an end-to-end AI pipeline for EEG-based medical decision support.

CERTIFICATIONS

- Generative - AI
- Programming Fundamentals (Python)
- IOT in Medical Applications
- Introduction to Internet Of Things (NPTEL)